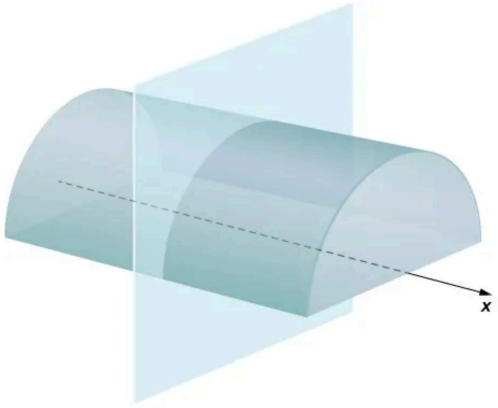
1 The Slicing Method

We now turn to using definite integrals to calculate the volume of three-dimensional solids. First, define a *cylinder* to be any solid that can be generated by translating a plane region (which is not necessarily a circle) along a line perpendicular to the region, called the *axis* of the cylinder. If we intersect any solid with a plane, then the intersection is called a *cross-section* of the solid. If the solid is a cylinder, then all cross-sections perpendicular to its axis are identical.



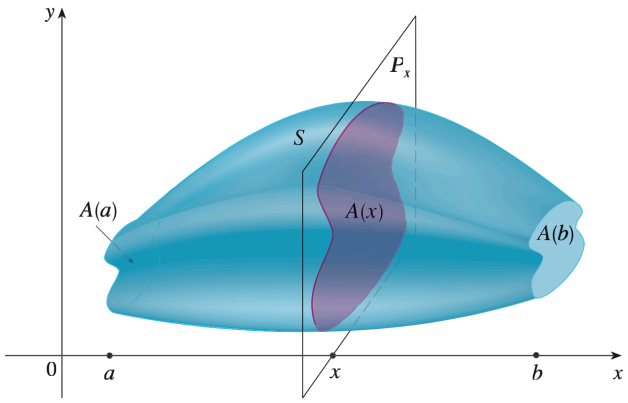
Three-dimensional cylinder



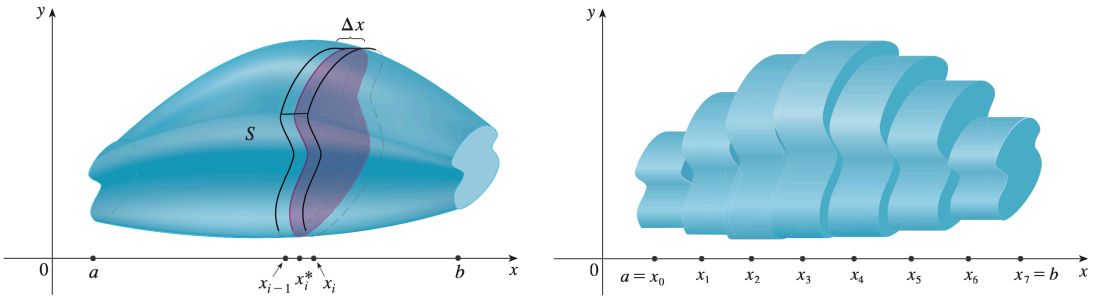
Two-dimensional cross section

To calculate the volume of a cylinder, we simply multiply the area of the cross-section by the height of the cylinder: $V = Ah$. For example, in the case of a right circular cylinder, this becomes $V = \pi r^2 h$.

Now consider a solid S which is not necessarily a cylinder. Let $A(x)$ be the area of the cross-section of S in a plane P_x perpendicular to the x-axis as passing through the point x, where $a \leq x \leq b$, as shown below.



The cross-sectional area $A(x)$ will vary as x increases from a to b. Partition the interval $[a,b]$ into n intervals of equal width $\Delta x = \frac{b-a}{n}$ and choose a tagging $\tau = \{x_i^*\}$ of this partition. Approximate the portion of the solid between x_{i-1} and x_i as a cylinder S_i with base area $A(x_i^*)$ and height Δx, which has volume $V(S_i) = A(x_i^*)\Delta x$.



The volume of the solid S is then approximately

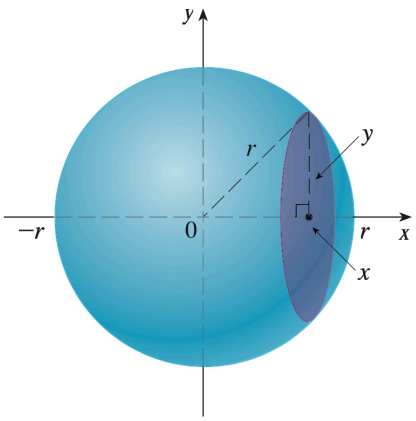
$$V \approx \sum_{i=1}^{n} A(x_i^*)\Delta x.$$

If the function $A(x)$ is integrable, then we define the volume of S to be

$$\boxed{V := \lim_{n\to\infty} A(x_i^*)\Delta x = \int_a^b A(x)dx}. \tag{1}$$

1. Consider a cylinder of base area A and height h. Choose the x axis to be the line perpendicular to the base. Show that Equation (1) gives $V = Ah$.

2. Show that the volume of a sphere of radius r is $V = \frac{4}{3}\pi r^3$.



2 Solids of Revolution and the Disk Method

If a region in the plane is revolved around a line in that plane, the resulting solid is called a *solid of revolution*. An example is shown in the figure below.

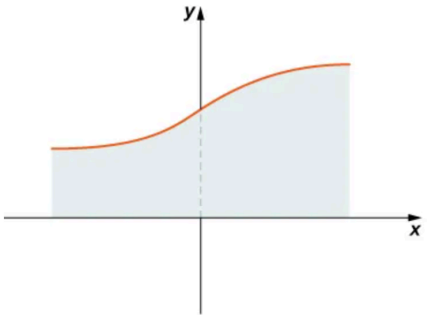


Figure 1: A region in the xy-plane.

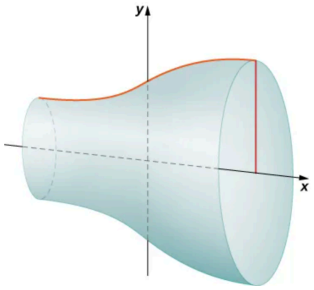


Figure 2: The solid of revolution produced by revolving the region in Figure 1 about the x-axis.

Solids of revolution are common in mechanical applications, such as machine parts produced by a lathe.

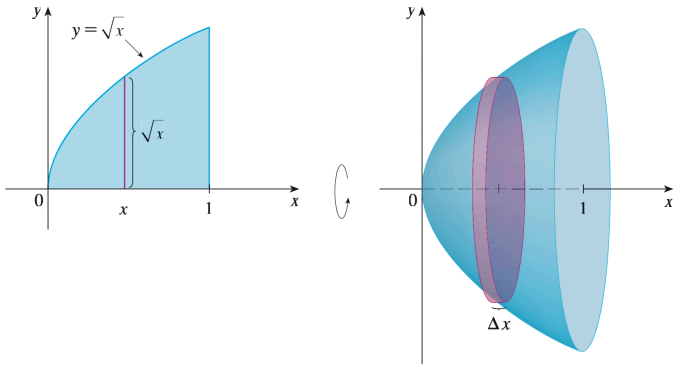
When we use the slicing method with solids of revolution, it is often called the *disk method* because each slice is a disk.

Theorem 2.1 (The Disk Method)**.** *Let $f(x)$ be a continuous, nonnegative function. Let R be the region abounded above by the graph of f, below by the x-axis, on the left by the line $x = a$, and on the right by the line $x = b$. Then, the volume of the solid of revolution formed by revolving R around the x-axis is given by*

$$\boxed{V = \int_a^b \pi(f(x))^2 dx.}$$

Proof. Since each cross-section is a disk of radius $f(x)$, $A(x) = \pi(f(x))^2$ and therefore the slicing method gives $V = \int_a^b A(x)dx = \int_a^b \pi(f(x))^2 dx$. □

3. Find the volume of the solid obtained by rotating about the x-axis the region under the curve $y = \sqrt{x}$ from 0 to 1.



4. Sketch the solid obtained by rotating the region between the graph of $f(x) = \sqrt{x}$ and the x-axis over the interval $[1,4]$ around the x-axis.

If R is a plane region bounded on the right by the graph of $g(y)$, on the left by the y-axis, below by the line $y = c$, and above by the line $y = d$, then the volume of the solid of revolution obtained by revolving R about the y-axis is $V = \int_c^d \pi(g(y))^2 dy$.

5. Sketch the solid obtained by rotating the region bounded by $y = x^3$, $y = 8$, and $x = 0$ about the y-axis, and calculate the volume of this region.